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Human Input Sensors for Shared Autonomy and HRC

Scope: A shorter, decision-oriented review of the main sensor modalities for human input in shared autonomy. The lists below are sorted by approximate price/performance value inside each category, from best value to premium.

How to read the shopping lists:

- Value rank is qualitative and combines cost, robustness, latency, and ease of deployment.
- Lower cost-per-useful-performance is better.
- Prices are typical academic/commercial street prices in 2024–2026 and can vary by vendor and bundle.
- For the final recommendation ranking, I weight **40% performance/integration**, **40% innovation**, and **20% price**.

1) Best Overall Recommendation

If you want the strongest practical stack for shared autonomy research, start with:

- **Hand pose** for continuous user intent and fine manipulation
- **Eye gaze** for attention and target disambiguation
- **Force / pressure** for contact and manipulation state
- **Voice** for high-level commands and safety overrides
- **Optional EMG** only if you specifically need fatigue, effort, or muscle-activation cues

This ordering reflects the weighted score above: strong real-time performance and integration, high novelty for shared autonomy, and still acceptable cost.

2) EMG: Muscle Activity

Best use: effort estimation, tremor/fatigue detection, grasp intent, assistive control research.

Value rank	Product	Typical price	Why it is in the list
1	MyoWare / low-cost sEMG kits	\$42.95	Cheapest way to prototype EMG signals
2	OpenBCI EMG-based setups	\$200–800	Flexible and good for research experiments
3	Delsys Trigno Wireless	\$40/sensor	Best overall research-grade choice

Short recommendation:

- Choose **MyoWare** if you want a cheap prototype.
- Choose **Delsys Trigno** if you want reliable research results.
- EMG is powerful but setup-heavy, so it is usually not the first sensor I would buy unless muscle intent is central to the thesis.

3) Eye Gaze: Attention and Target Selection

Best use: target inference, object selection, shared autonomy arbitration.

Value rank	Product	Typical price	Why it is in the list
1	GazePoint GP3	\$985.00	Lowest-cost usable research eye tracker

Value rank	Product	Typical price	Why it is in the list
2	Pupil Core	\$3,000–3,500	Strong value for wearable and lab work
3	Tobii Pro Nano	\$3,500–4,500	Best balance of accuracy, ease, and stability

Short recommendation:

- **Tobii Pro Nano** is the best default choice for shared autonomy.
- **Pupil Core** is the better budget-research option if you want head-mounted flexibility.
- Eye gaze is one of the most valuable modalities because it directly captures attention before motion happens.

4) Motion Capture / Skeletal Tracking

Best use: full-body pose, arm motion, gesture tracking, teleoperation, training data.

Value rank	Product	Typical price	Why it is in the list
1	MediaPipe Pose / Hands + webcam	\$0–200	Best cost/performance for quick prototyping
2	Intel RealSense	\$400	Low-cost depth-based skeletal tracking

Short recommendation:

- Start with **MediaPipe** if you need a fast and cheap baseline.
- Use **Azure Kinect** or **RealSense** if you need depth-based body pose.
- Use **Vicon** only when accuracy is more important than budget.

5) BCI / EEG

Best use: special populations, research on non-muscular control, niche assistive applications.

Value rank	Product	Typical price	Why it is in the list
1	OpenBCI Cyton	\$1250	Lowest-cost serious EEG research platform
2	Emotiv EPOC X	\$1,199	Easy to buy and quick to start

Short recommendation:

- EEG is innovative, but the cost/performance is usually worse than gaze, hand pose, or force for shared autonomy.
- I would only choose EEG if you need a strong novelty angle or must work with users who cannot provide conventional motor input.

6) Voice / Speech

Best use: high-level commands, corrections, emergency stop, low-bandwidth context.

Value rank	Product	Typical price	Why it is in the list
1	Vosk / Silero / Whisper local	open source	Cheapest and privacy-friendly if you can run locally
2	Azure Speech	pay as you go	Good accuracy and easy integration
3	Google Speech-to-Text	pay as you go	Strong accuracy, cloud dependence
4	AWS Transcribe / premium cloud stacks	pay as you go	Strong but usually not needed first

Short recommendation:

- Local speech models are the best value if you can tolerate some tuning.

- Cloud speech is better when you want fast deployment and don't mind network dependence.

7) Hand Pose / Hand Tracking

Best use: grasp intention, finger gestures, direct manipulation cues, shared autonomy.

Value rank	Product	Typical price	Why it is in the list
1	MediaPipe Hands	\$0 + webcam	Best value by far for real-time hand tracking
2	Leap Motion Controller 2 / Ultraleap	\$285	Very good precision and easy setup
3	Rokoko smart gloves	\$1795	Premium, specialized, and expensive

Short recommendation:

- **MediaPipe Hands** is the best overall value for shared autonomy.
- **Leap Motion / Ultraleap** is the best low-cost hardware option if you want dedicated hand sensing.
- This category is one of the strongest for your thesis because it is cheap, real-time, and very expressive.